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Glove

Abstract

A glove including a top surface, a bottom surface opposite the top surface, and the bottom surface is configured to interface with an object being gripped by a user. The glove also includes an index finger portion configured to receive an index finger of the user. The index finger portion includes a knuckle configured to pivot about an axis. The glove further includes a gripping portion coupled to the index finger portion between the top and bottom surfaces. The gripping portion includes a first notch adjacent the top surface and a second notch adjacent the bottom surface. The first notch defines a first angle, and the second notch defines a second angle, wherein the first angle increases and the second angle decreases as the knuckle pivots about the axis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 15/046,597, filed Feb. 18, 2016, which claims priority to U.S. Provisional Application No. 62/117,622, filed Feb. 18, 2015, and to U.S. Provisional Application No. 62/234,131, filed Sep. 29, 2015, which are incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

(1) The present invention relates to gloves, and more particularly to work gloves.

SUMMARY

(2) In one aspect, the invention provides a glove including a top surface and a bottom surface opposite the top surface with the bottom surface configured to interface with an object being gripped by a user. The glove also includes an index finger portion configured to receive an index finger of the user. The index finger portion includes a knuckle configured to pivot about an axis.

The glove further includes a gripping portion coupled to the index finger portion between the top and bottom surfaces. The gripping portion includes a first notch adjacent the top surface and a second notch adjacent the bottom surface. The first notch defines a first angle, and the second notch defines a second angle, wherein the first angle increases and the second angle decreases as the knuckle pivots about the axis.

(3) In another aspect, the invention provides a glove including a top surface and a bottom surface opposite the top surface with the bottom surface configured to interface with an object being gripped by a user. The glove also includes a thumb portion configured to receive a thumb of the user and an index finger portion configured to receive an index finger of the user. The index finger portion includes a knuckle configured to pivot about an axis. The glove further includes a gripping portion coupled to the thumb portion and the index finger portion. The gripping portion includes a first notch adjacent the top surface and a second notch adjacent the bottom surface. The first and second notches are associated with the axis of the index finger portion. The glove includes a capacitive material coupled to the knuckle of the index finger portion. The capacitive material is configured to bend about the axis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a top perspective view of a right hand glove according to an embodiment of the invention.

(2) FIG. 2 is a top view of the right hand glove of FIG. 1.

(3) FIG. 3 is a bottom view of the right hand glove of FIG. 1.

(4) FIG. 4 is a left side view of the right hand glove of FIG. 1.

(5) FIG. 5 is a right side view of the right hand glove of FIG. 1.

(6) FIG. 6 is a front view of the right hand glove of FIG. 1.

(7) FIG. 7 is a rear view of the right hand glove of FIG. 1.

(8) FIG. 8 is a detailed view of FIG. 4 illustrating a portion of an index finger of the right hand glove being bent.

(9) Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

(10) FIGS. 1-8 illustrate a glove 10 (e.g., a work glove) according to one embodiment. The illustrated glove 10 includes a top surface 14 opposite a bottom surface 18 and a palm portion 20 associated with the bottom surface 18. The illustrated bottom surface 18 is configured to interface with an object being gripped. In the illustrated embodiment, the bottom surface 18 is substantially made from synthetic leather, and the top surface 14 is substantially made from spandex. In other embodiments, the surfaces 14, 18 may be of different material.

(11) A thumb portion 26, an index finger portion 30, a middle finger portion 34, a ring finger portion 38, and a little finger portion 42 are configured to receive a respective finger of a user's hand after the user's hand is inserted through a cuff 45. In other embodiments, the portions 26, 30, 34, 38, 42 may not include a closed end or tip (e.g., may include an open end) such that the user's thumb and fingers extend through and beyond the portions 26, 30, 34, 38, 42 (e.g., a framer or fingerless glove design). The illustrated cuff 45 includes a strap 46 that is removably attached to the cuff 45 by a Velcro® closure, thereby allowing the cuff 45 to be adjusted to various sized wrists of

a user's hand. In other embodiments, the strap **46** is replaced with an elastic band that forms over a wrist of the user's hand. A tab **47** is also attached to a portion of the cuff **45** adjacent the bottom surface portion **18** to aid in pulling the glove **10** over the user's hand. In the illustrated embodiment, the strap **46** is manufactured from rubber, but in other embodiments the strap **46** may be manufactured from a fabric material.

(12) The illustrated index finger portion **30** includes a first knuckle **44** and a second knuckle **48** with the second knuckle **48** located at a base of the index finger portion **30** away from the end or tip of the index finger portion **30**. The first knuckle **44** is located between the second knuckle **48** and the tip of the index finger portion **30**. The first knuckle **44** generally pivots about an axis **50**, for example, when a user bends their index finger. Although not illustrated, the finger portions **34**, **38**, **42** also pivot about a similar axis to the axis **50**, and in other embodiments, a slot or break in material is associated with the axis of the finger portions **34**, **38**, **42**. For example, an elastic material may form a portion of the finger portions **34**, **38**, **42** that bends about the axis.

(13) In the illustrated embodiment, a capacitive material **64** is associated with the first knuckle **44**. In particular, a midpoint of the capacitive material **64** is configured to bend about the axis **50**. In other embodiments, the axis **50** may be offset relative to the midpoint of the capacitive material **64**. The illustrated capacitive material **64** includes conductive particles (e.g., graphite) penetrated into a synthetic material (e.g., composition of spandex and polyurethane). In other embodiments, the capacitive material **64** may include different types of materials. The capacitive material **64** is configured to operate a capacitive sensing electronic device (e.g., a smartphone) by transferring the capacitance of the user's finger to the electronic device. In other embodiments, the capacitive material **64** may be located on another knuckle of another finger portion **34**, **38**, **42**.

(14) A side gripping portion **54** is coupled between the thumb portion **26** and the index finger portion **30** and also extends between the top surface **14** and the bottom surface **18**. In other words, the side gripping portion **54** is generally located on a side of the thumb portion **26** and a side of the index finger portion **30**. The illustrated gripping portion **54** provides an increased coefficient of friction between the glove **10** and an object being gripped to reduce slip therebetween. In the illustrated embodiment, the gripping portion **54** is manufactured from an anti-slip rubber compound (e.g., Armortex® (NL-8050)), which includes greater rigidity than the surrounding material of the index finger portion **30**. In other embodiments, the side gripping portion **54** may be of different material that is more rigid than the material of the index finger portion **30**. The illustrated gripping portion **54** includes flexible notches **62** with a first notch **62a** adjacent the top surface **14** and a second notch **62b** opening opposite from the first notch **62a** and positioned adjacent the bottom surface **18**. In other embodiments, the gripping portion **54** may only include one notch **62**. An apex **63** of each notch **62** is curvilinear or U-shaped (FIG. 4); however, in other embodiments, the apex **63** may be V-shaped. Each apex **63** of the illustrated notches **62** is located on a side surface **65** (FIG. 8) of the first knuckle **44** of the index finger portion **30** (the side surface **65** is positioned between the top surface **14** and the bottom surface **18**) so that the notches **62** generally align with the axis **50**. In particular, the axis **50** is positioned between the apexes **63** of the notches **62**. In addition, first ends **67** and second ends **68** of the notches **62a**, **62b** are located on the side surface **65** of the first knuckle **44** (FIG. 8).

(15) With reference to FIGS. 3-7, the bottom surface **18** includes bottom gripping portions **58**, similar to the side gripping portion **54**. In one embodiment, the gripping portions **58** can also include padding or cushioning material to decrease vibrational forces transmitted to the glove **10** and to improve comfort of wearing the glove **10**. The bottom gripping portions **58** are located on the palm portion **20**, at ends of the thumb and finger portions **26**, **30**, **34**, **38**, **42**, and at an area between the palm portion **20** and the finger portions **30**, **34**, **38**, **42**.

(16) With reference to FIGS. 1-5 and 7, the illustrated glove **10** also includes an absorbent portion **66** extending from the cuff **45** along a side of the thumb portion **26** and also extending between the top surface **14** and the bottom surface **18**. In other words, the absorbent portion **66** defines a portion

of the thumb portion **26**. The illustrated absorbent portion **66** includes a loop material (e.g., a terrycloth material) configured to absorb moisture. For example, as the absorbent portion **66** contacts skin of the user (e.g., a user's forehead) the moisture thereon is absorbed by the absorbent portion **66**. In other embodiments, the absorbent portion **66** may be of a different absorbent material.

(17) With reference to FIGS. **1**, **2**, **4** and **5**, a guard **74** is coupled to the top surface **14** and is positioned at a base of the finger portions **30**, **34**, **38**, **42**. The illustrated guard is configured to protect the user's hand. The illustrated guard **74** includes protrusions **78** with one protrusion extending towards the middle finger portion **34** and another protrusion **78** extending towards the ring finger portion **38**. In other embodiments, the guard **74** may include more or less than two protrusions **78**, and/or the protrusions **78** may extend towards other finger portions. Line features **82** defined with contrasting color to the guard **74** extend between the index finger portion **30** and the little finger portion **42**. For example, the line features **82** are red in color (e.g., Pantone 18-1763 TCX), whereas the guard **74** is substantially black in color. In addition, an accent feature **84** is positioned on the top surface **14** and extends from the index finger portion **30** to the cuff **45** and partially surrounds the capacitive material **64**. The illustrated accent feature **84** also abuts the side gripping portion **54** and the absorbent portion **66**. The accent feature **84** is illustrated in contrasting color with bordering portions of the glove **10**. For example, the accent feature **84** is red in color (e.g., Pantone 18-1763 TCX) with the bordering areas generally black or grey in color. In other embodiments, an accent stitching (e.g., red stitching) may surround the capacitive material **64**. In further embodiments, the accent feature **84** may be positioned and oriented differently on the top surface **14** of the glove **10** (e.g., the accent feature **84** may extend between the middle finger portion **34** and the little finger portion **42**).

(18) In operation, as the user's hand is within the glove **10** and the user's index finger is oriented straight (e.g., not bent), the notch **62a** defines a first angle **86** and the notch **62b** defines a second angle **90** with the angles **86**, **90** being substantially the same (FIG. **4**). As the user bends or pivots their index finger about the axis **50**, the first knuckle **44** also pivots about the axis **50** thereby increasing the first angle **86** and decreasing the second angle **90** (FIG. **8**). Because the side gripping portion **54** is more rigid than other portions of the index finger portion **30**, the notches **62** are flexible so that the side gripping portion **54** does not impede (e.g., reduce resistance) bending of the first knuckle **44**. As the user bends their index finger, the capacitive material **64** and the notches **62** concurrently bend about the axis **50** to extend their knuckle (e.g., the first knuckle **44**) to operate the electric device by the capacitive material **64**.

(19) Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described.

Claims

1. A work glove comprising: a top surface; a bottom surface opposite the top surface; a middle finger portion configured to receive a middle finger of a user; a ring finger portion configured to receive a ring finger of a user; a little finger portion configured to receive a little finger of a user; an index finger portion having a first material and configured to receive an index finger of the user, the index finger portion comprising a side positioned between the top surface and the bottom surface, the index finger portion comprising: a first knuckle portion; a second knuckle portion located at a base of the index finger portion, wherein the first knuckle portion is located between the second knuckle portion and a tip of the index finger portion and configured to bend about an axis; and a thumb portion configured to receive a thumb of the user, the thumb portion comprising a side between the top surface and the bottom surface and facing the side of the index finger portion; a side gripping portion extending along the side of the index finger portion to the first

knuckle portion, the side gripping portion comprising: a second material having a coefficient of friction greater than a coefficient of friction of the first material and a rigidity greater than a rigidity of the first material; a first notch defined in the side gripping portion facing the top surface and comprising an apex located on the side of the index finger at the first knuckle portion, wherein the first notch is aligned with the axis of the first knuckle portion; and a second notch defined in the side gripping portion facing the bottom surface and comprising an apex located on the side of the index finger at the first knuckle portion, wherein the second notch is aligned with the axis of the first knuckle portion; a bottom gripping portion located on the bottom surface; and a guard portion coupled to the top surface and located at bases of the index finger portion, the middle finger portion, the ring finger portion and the little finger portion.

2. The work glove of claim 1, further comprising a capacitive material located at a top surface of the first knuckle portion of the index finger portion, wherein the capacitive material is configured to operate a capacitive sensing electronic device, and wherein the capacitive material is positioned to align with the first and second notches of the side gripping portion such that the capacitive material is configured to bend about the axis associated with the first knuckle portion of the index finger.

3. The work glove of claim 2, wherein the bottom surface comprises a synthetic leather material, the top surface comprises a spandex material and the side gripping portion comprises an anti-slip rubber compound.

4. The work glove of claim 3, further comprising an absorbent portion extending along a side of the thumb portion, the absorbent portion comprising a loop material configured to absorb moisture from a user's skin.

5. The work glove of claim 1, wherein the bottom gripping portion comprises a plurality of discrete bottom gripping portions, wherein at least one of the discrete bottom gripping portions is located on each of a palm portion, an end of the index finger portion, an end of the middle finger portion, an end of the ring finger portion, an end of the little finger portion and an area between the palm portion and the index finger portion, middle finger portion, ring finger portion and little finger portion.

6. The work glove of claim 5, wherein each of the plurality discrete bottom gripping portions comprises a padding material configured to decrease transmission of vibrational forces to a hand of the user.

7. The work glove of claim 1, wherein the guard portion is a contiguous guard portion extending across bases of the index finger portion, the middle finger portion, the ring finger portion and the little finger portion.

8. The work glove of claim 7, wherein the guard portion further comprises a first protrusion extending toward middle finger portion, and a second protrusion extending toward the ring finger portion.

9. The work glove of claim 1, wherein the side gripping portion is a contiguous side gripping portion extending from the side of the thumb portion to the side of the first knuckle portion of the index finger portion.

10. The work glove of claim 1, wherein the apex of the first notch is aligned with the axis of the first knuckle portion and the apex of the second notch is aligned with the axis of the first knuckle portion, wherein the axis is positioned between the apex of the first and second notches.

11. The work glove of claim 10, wherein an edge of the side gripping portion that defines the first notch has a curvilinear shape, wherein an edge of the side gripping portion that defines the second notch has a curvilinear shape.

12. The work glove of claim 11, wherein the first and second notches are symmetrical about the axis of the first knuckle portion.

13. A work glove comprising: a top surface; a bottom surface opposite the top surface; a middle finger portion configured to receive a middle finger of a user; a ring finger portion configured to

receive a ring finger of a user; a little finger portion configured to receive a little finger of a user; an index finger portion having a first material and configured to receive an index finger of the user, the index finger portion comprising a side positioned between the top surface and the bottom surface, the index finger portion comprising: a first knuckle portion; a second knuckle portion located at a base of the index finger portion, wherein the first knuckle portion is located between the second knuckle portion and a tip of the index finger portion and configured to bend about an axis; and a thumb portion configured to receive a thumb of the user, the thumb portion comprising a side between the top surface and the bottom surface and facing the side of the index finger portion; a side gripping portion on the side of the index finger portion at the first knuckle portion, the side gripping portion comprising: a second material different than the first material; a first notch defined in the side gripping portion facing the top surface and comprising an apex located on the side of the index finger at the first knuckle portion; and a second notch defined in the side gripping portion facing the bottom surface and comprising an apex located on the side of the index finger at the first knuckle portion; and a capacitive material located at a top surface of the first knuckle portion of the index finger portion, wherein the capacitive material is configured to operate a capacitive sensing electronic device, and wherein the capacitive material is positioned to align with the first and second notches of the side gripping portion such that the capacitive material is configured to bend about the axis associated with the first knuckle portion of the index finger; wherein the bottom surface comprises a synthetic leather material, the top surface comprises a spandex material and the side gripping portion comprises an anti-slip rubber compound.
